

100 DSA Questions for Logic Building

How to Think of Logic

Break Down the Problem: Read the question twice. Identify inputs, outputs, and constraints.

Start with Examples: Create test cases manually and simulate the solution step-by-step.

Write Pseudocode: Draft a high-level plan before coding.

Ask “Why?” at Every Step: Understand each operation; don’t memorize solutions.

Visualize: Use diagrams or dry-run tables to debug.

Optimize Gradually: Start with brute force; iterate to optimize for efficiency.

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Category	Question Name	Link	Platform
Basics & Warm-Up	Print a pattern of stars (triangle, pyramid)	Link	GeeksforGeeks
	Reverse a number	Link	GeeksforGeeks
	Check if a number is palindrome	Link	GeeksforGeeks
	Count digits in a number	Link	GeeksforGeeks

Arrays	Find factorial of a number	Link	HackerRank
	Reverse an array	Link	LeetCode
	Find the maximum and minimum of an array	Link	GeeksforGeeks
	Rotate an array by K steps	Link	LeetCode
	Move all zeroes to the end	Link	LeetCode
Strings	Kadane's Algorithm (Maximum Subarray Sum)	Link	LeetCode
	Reverse a string	Link	LeetCode
	Check if two strings are anagrams	Link	LeetCode
	Longest Common Prefix	Link	LeetCode
	Check if a string is a palindrome	Link	LeetCode
Recursion	Count and say	Link	LeetCode
	Fibonacci series using recursion	Link	GeeksforGeeks
	Tower of Hanoi	Link	GeeksforGeeks
	Factorial using recursion	Link	LeetCode
	Reverse a linked list using recursion	Link	LeetCode
	Permutations of a string	Link	GeeksforGeeks

Sorting	Bubble sort	Link	GeeksforGeeks
	Selection sort	Link	GeeksforGeeks
	Merge sort	Link	GeeksforGeeks
	Quick sort	Link	GeeksforGeeks
	Insertion sort	Link	GeeksforGeeks
Searching	Binary search	Link	LeetCode
	Linear search	Link	GeeksforGeeks
	Search in a rotated sorted array	Link	LeetCode
	First and last position in a sorted array	Link	LeetCode
	Square root of a number (using binary search)	Link	LeetCode
Linked List	Reverse a linked list	Link	LeetCode
	Detect a cycle in a linked list	Link	LeetCode
	Merge two sorted linked lists	Link	LeetCode
	Remove Nth node from the end	Link	LeetCode
	Find the middle of a linked list	Link	LeetCode
Stacks & Queues	Implement a stack using arrays	Link	LeetCode

Dynamic Programming	Evaluate postfix expression	Link	GeeksforGeeks
	Balanced parentheses	Link	LeetCode
	Next greater element	Link	LeetCode
	Implement a queue using stacks	Link	LeetCode
	0/1 Knapsack problem	Link	GeeksforGeeks
	Fibonacci using dynamic programming	Link	LeetCode
	Longest common subsequence	Link	LeetCode
	Longest increasing subsequence	Link	LeetCode
Graphs	Minimum steps to reach the end	Link	LeetCode
	BFS traversal	Link	GeeksforGeeks
	DFS traversal	Link	GeeksforGeeks
	Detect cycle in an undirected graph	Link	GeeksforGeeks
	Shortest path in a graph (Dijkstra's)	Link	GeeksforGeeks
	Topological sort	Link	GeeksforGeeks